Vector Spaces, Subspaces, Span, and Bases

Linear Algebra · Lecture 3

Last time (Lecture 2): row operations take $[A \mid b]$ to an RREF, from which consistency and all solutions are read off (pivot and free unknowns); a square A is invertible $\iff$ $Ax = 0$ only for $x = 0$.

From Lecture 1 and Lecture 2 (restated; $A \in \mathbb{R}^{m\times n}$ has the columns $a_1, \dots, a_n$).

- Lecture 1, Definition 1, Theorem 3(a),(b): $A + B$, cA are taken entry by entry and obey the laws of numbers; $-A = (-1)A$; 0 is the zero matrix; $e_j = \mathrm{col}_j(I_n)$. (a) $A(BG) = (AB)G$, $A(B+G) = AB+AG$, $(A+B)G = AG+BG$, $A(cB) = c(AB)$, $I_mA = A = AI_n$, $0A = 0$, $A0 = 0$. (b) A 1×1 factor c acts as a number: $Mc = cM$.
- Lecture 1, Theorem 4(a),(b): $\mathrm{col}_j(AB) = A\,\mathrm{col}_j(B)$; $Ax = x_1a_1 + \cdots + x_na_n$ ($x \in \mathbb{R}^n$), $Ae_j = a_j$.
- Lecture 1, Definition 7, Theorem 8: $(A')_{ij} = a_{ji}$; A is **symmetric** if $A' = A$; $(A+B)' = A' + B'$, $(cA)' = cA'$, $(AB)' = B'A'$.
- Lecture 1, Definition 14, Proposition 15, Example 16: $\mathbf{1} = \mathbf{1}_n = (1, \dots, 1)' \in \mathbb{R}^n$, $\bar{J}_n = \frac{1}{n}\mathbf{1}\mathbf{1}'$, $C = C_n = I_n - \bar{J}_n$; $\mathbf{1}'\mathbf{1} = n$, $C' = C$, $C\mathbf{1} = 0$; for $x \in \mathbb{R}^n$, $\bar{x} = \frac{1}{n}\mathbf{1}'x$ and $Cx = x - \bar{x}\mathbf{1}$.
- Lecture 2, Theorem 3(b): if row operations take $[A \mid b]$ to $[B \mid z]$, then $Ax = b$ and $Bx = z$ have the same solutions.
- Lecture 2, Definitions 1, 4: $Ax = b$ is **consistent** if it has a solution. R is in **RREF** if the first nonzero entry of each nonzero row is 1 (its **pivot**), the only nonzero entry of its column; zero rows are last; the pivots of rows $1, \dots, r$ lie in columns $k_1 < \cdots < k_r$ (the **pivot columns**; $\mathrm{col}_{k_i}(R) = e_i$). E.g. I_n.
- Lecture 2, Proposition 7: let row operations take $[A \mid b]$ to $[R \mid z]$, R in RREF with nonzero rows $1, \dots, r$ and pivot columns $k_1 < \cdots < k_r$; $x_{k_1}, \dots, x_{k_r}$ are the **pivot unknowns**, the others the **free unknowns**. (a) $Ax = b$ is consistent $\iff$ $z_i = 0$ for all $i > r$. (b) If so, the solutions are: free unknowns arbitrary, $x_{k_i} = z_i - \sum_{j \text{ free}} R_{ij}x_j$ ($i \leqslant r$).
- Lecture 2, Theorems 9(a), 22(b): if $m < n$, then $Ax = 0$ for some $x \neq 0$; a consistent $Ax = b$ has exactly one solution $\iff$ it has no free unknowns.
- Lecture 2, Theorem 14(c): if row operations $\omega_1, \dots, \omega_k$ take A to B, then $B = PA$ with $P = E_k \cdots E_1$, $E_j = \omega_j(I_m)$, invertible. P depends only on the operations, and $\mathrm{col}_j(B) = Pa_j$ (Lecture 1, Theorem 4(a)): the same operations take any columns of A to the corresponding columns of B.
- Lecture 2, Definition 10, Theorem 15: $A \in \mathbb{R}^{n\times n}$ is **invertible** if $AB = BA = I_n$ for some B, the inverse A^{-1}; A is invertible $\iff$ $(Ax = 0 \Rightarrow x = 0)$.
- Lecture 2, Example 6: row operations take $A = \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & 2 & -3 \\ 2 & 1 & 5 & -4 \end{pmatrix}$ to $R = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \end{pmatrix}$, in RREF; $Ax = 0 \iff x = t(-2, 3, 1, 1)'$ ($t \in \mathbb{R}$).
- Lecture 2, Examples 19, 23: $A = \begin{pmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -2 & 1 \\ 2 & -1 & 2 & -1 \\ 1 & -1 & 1 & 0 \end{pmatrix}$ is invertible, $A^{-1} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 1 & -1 \\ -1 & -1 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}$; $Ax = (1, 0, 3, 3)'$ has the solution $x = (3, -1, -1, 2)'$.

1. Vector spaces and subspaces

Definition 1. A (real) **vector space** is a set V (its elements are **vectors**) with an **addition** $u+v \in V$ ($u, v \in V$) and a **scalar multiplication** $cv \in V$ ($c \in \mathbb{R}$, $v \in V$) such that for all $u, v, w \in V$, $c, d \in \mathbb{R}$: (V1) $u+v = v+u$; (V2) $u+(v+w) = (u+v)+w$; (V3) there is a **zero vector** $0 \in V$ with $v+0 = v$ for all $v \in V$; (V4) each $v \in V$ has a $-v \in V$ with $v + (-v) = 0$ ($u - v = u + (-v)$); (V5) $1v = v$; (V6) $(cd)v = c(dv)$; (V7) $c(u+v) = cu + cv$; (V8) $(c+d)v = cv + dv$. A vector $c_1v_1 + \cdots + c_kv_k$ ($c_i \in \mathbb{R}$) is a **linear combination** of $v_1, \dots, v_k$. $\mathbb{R}^n$ and $\mathbb{R}^{m\times n}$, with the entrywise $A+B$, cA, are vector spaces: (V1)–(V8) hold entry by entry, the zero vector is the zero matrix, and $-A = (-1)A$ (Lecture 1, Definition 1).

Proposition 2. Let V be a vector space, $v \in V$, $c \in \mathbb{R}$. Then $0v = 0$, $(-1)v = -v$, $c0 = 0$; and $cv = 0 \Rightarrow c = 0$ or $v = 0$.

Proof.

Add $-(0v)$, $-(c0)$ to both sides in lines 1, 4; line 3: any w with $v + w = 0$ is $-v$, so $(-1)v = -v$ (line 2):

$$0v = (0+0)v = 0v + 0v \implies 0 = 0v \qquad \text{((V8); (V2)–(V4))}$$

$$v + (-1)v = 1v + (-1)v = \Big(1 + (-1)\Big)v = 0v = 0 \qquad \text{((V5), (V8); line 1)}$$

$$w = w + \Big(v + (-v)\Big) = (w + v) + (-v) = 0 + (-v) = -v \qquad \text{((V1)–(V4))}$$

$$c0 = c(0+0) = c0 + c0 \implies 0 = c0 \qquad \text{((V3), (V7); (V2)–(V4))}$$

$$c \neq 0,\ cv = 0 \implies v = 1v = (c^{-1}c)v = c^{-1}(cv) = c^{-1}0 = 0 \qquad \text{((V5), (V6); line 4)}$$

□

Definition 3. A **subspace** of a vector space V is a subset $W \subseteq V$ that is a vector space with the addition and the scalar multiplication of V.

Theorem 4. For a subset W of a vector space V, the following are equivalent:

(a) W is a subspace of V;

(b) $0 \in W$, and $u + v \in W$, $cu \in W$ for all $u, v \in W$, $c \in \mathbb{R}$ (W is **closed** under addition and scalar multiplication);

(c) $0 \in W$, and $cu + v \in W$ for all $u, v \in W$, $c \in \mathbb{R}$.

Proof.

(a)⇒(b)⇒(c):

$$\text{(a)} \implies u + v \in W,\ cu \in W\ (u, v \in W,\ c \in \mathbb{R}) \qquad \text{(the operations of } V\text{)}$$

$$\text{(a)} \implies \text{some } w \in W\ (\text{(V3) in } W) \implies 0 = 0w \in W \qquad \text{(line 1; Proposition 2)}$$

$$\text{(b)} \implies cu \in W \implies cu + v \in W$$

(c)⇒(a): the operations of V give operations on W, and (V1)–(V8) hold:

$$v = 0:\ cu = cu + 0 \in W; \qquad c = 1:\ u + v = 1u + v \in W \qquad \text{((V3), (V5))}$$

(V1), (V2), (V5)–(V8) hold in W, as they hold in V; (V3): $0 \in W$

$$\text{(V4)}:\ -w = (-1)w \in W\ (w \in W) \qquad \text{(Proposition 2)}$$

□

So W is a subspace if it contains 0 and is closed under the two operations; it is not, if one of these fails.

Example 5. (a) The symmetric $n \times n$ matrices form a subspace of $\mathbb{R}^{n\times n}$.

(b) For $A \in \mathbb{R}^{m\times n}$, $\{x \in \mathbb{R}^n : Ax = 0\}$ is a subspace of $\mathbb{R}^n$; for $b \neq 0$, $\{x \in \mathbb{R}^n : Ax = b\}$ is not.

(c) The **centered** vectors $W_0 = \{x \in \mathbb{R}^n : \mathbf{1}'x = 0\}$ form a subspace of $\mathbb{R}^n$, and $Cx \in W_0$ for every $x \in \mathbb{R}^n$.

(d) $\{(x, y, z)' \in \mathbb{R}^3 : xy = 0\}$, the union of the subspaces $\{x = 0\}$ and $\{y = 0\}$, is not a subspace of $\mathbb{R}^3$.

Solution.

(a) Theorem 4(c); A, B symmetric:

$$0' = 0, \qquad (cA + B)' = cA' + B' = cA + B \qquad \text{(Lecture 1, Theorem 8)}$$

(b) Theorem 4(c); $Au = Av = 0$:

$$A0 = 0, \qquad A(cu + v) = cAu + Av = 0 \qquad \text{(Lecture 1, Theorem 3(a))}$$
$$A0 = 0 \neq b \implies 0 \notin \{x : Ax = b\} \qquad \text{(Theorem 4(b))}$$

(c) W_0 is (b) with the $1 \times n$ matrix $A = \mathbf{1}'$; $C' = C$, $C\mathbf{1} = 0$ (Lecture 1, Proposition 15):

$$\mathbf{1}'Cx = \mathbf{1}'C'x = (C\mathbf{1})'x = 0'x = 0 \qquad \text{(Lecture 1, Theorem 8)}$$

(d) $\{x = 0\}$, $\{y = 0\}$ are (b) with $A = (1\ 0\ 0)$, $(0\ 1\ 0)$; Theorem 4(b) fails:

$(1, 0, 0)'$, $(0, 1, 0)'$ lie in the set $(1 \cdot 0 = 0 \cdot 1 = 0)$; $\quad (1, 0, 0)' + (0, 1, 0)' = (1, 1, 0)'$ does not $(1 \cdot 1 \neq 0)$
So a union of subspaces need not be a subspace.

2. Span

Definition 6. For $v_1, \ldots, v_k$ in a vector space V: $\operatorname{span}\{v_1, \ldots, v_k\} = \{c_1v_1 + \cdots + c_kv_k : c_1, \ldots, c_k \in \mathbb{R}\}$, the **span** of $v_1, \ldots, v_k$; $\operatorname{span} \varnothing = \{0\}$. If $\operatorname{span}\{v_1, \ldots, v_k\} = W$, then $v_1, \ldots, v_k$ **span** W.

Theorem 7. $\operatorname{span}\{v_1, \ldots, v_k\}$ is a subspace of V that contains $v_1, \ldots, v_k$, and it is contained in every subspace of V that contains $v_1, \ldots, v_k$: it is the smallest such subspace.

Proof.

$k = 0$ and $k \geqslant 1$:

$$k = 0: \ \operatorname{span} \varnothing = \{0\} \ni c0 + 0 = 0; \ 0 \in \text{every subspace} \qquad \text{(Proposition 2; Theorem 4)}$$
$$k \geqslant 1: \ 0 = 0v_1 + \cdots + 0v_k, \quad c\textstyle\sum_i a_iv_i + \sum_i b_iv_i = \sum_i (ca_i + b_i)v_i \qquad \text{(Theorem 4(c))}$$
$$v_j = 1v_j + \textstyle\sum_{i\neq j} 0v_i \in \operatorname{span}\{v_1, \ldots, v_k\} \qquad \text{(Proposition 2)}$$
$$U \ni v_1, \ldots, v_k \text{ a subspace} \implies c_iv_i \in U, \ c_1v_1 + \cdots + c_kv_k \in U \qquad \text{(Theorem 4(b))}$$

□

For $v_1, \ldots, v_k \in \mathbb{R}^m$ and $V = [v_1 \ \cdots \ v_k] \in \mathbb{R}^{m\times k}$, $Vc = c_1v_1 + \cdots + c_kv_k$ (Lecture 1, Theorem 4(b)), so

$$\operatorname{span}\{v_1, \ldots, v_k\} = \{Vc : c \in \mathbb{R}^k\}, \qquad b \in \operatorname{span}\{v_1, \ldots, v_k\} \iff Vc = b \text{ is consistent}$$

and this is decided by row-reducing $[V \mid b]$ (Lecture 2, Proposition 7).

Example 8. Let $W = \operatorname{span}\{v_1, v_2, v_3\} \subseteq \mathbb{R}^5$, $v_1 = (1,2,0,3,0)'$, $v_2 = (0,0,1,4,0)'$, $v_3 = (0,0,0,0,1)'$. Which $y = (y_1, \ldots, y_5)'$ lie in W? Do $(-3,-6,1,-5,2)'$ and $(2,4,6,7,8)'$?

Solution.

Row-reduce $[v_1\ v_2\ v_3 \mid y]$, keeping the y_i as letters:

$$\left(\begin{array}{ccc|c} 1&0&0&y_1\\ 2&0&0&y_2\\ 0&1&0&y_3\\ 3&4&0&y_4\\ 0&0&1&y_5 \end{array}\right) \xrightarrow[R_4\leftarrow R_4-3R_1]{R_2\leftarrow R_2-2R_1} \left(\begin{array}{ccc|c} 1&0&0&y_1\\ 0&0&0&y_2-2y_1\\ 0&1&0&y_3\\ 0&4&0&y_4-3y_1\\ 0&0&1&y_5 \end{array}\right) \xrightarrow{R_4\leftarrow R_4-4R_3} \left(\begin{array}{ccc|c} 1&0&0&y_1\\ 0&0&0&y_2-2y_1\\ 0&1&0&y_3\\ 0&0&0&y_4-3y_1-4y_3\\ 0&0&1&y_5 \end{array}\right)$$

$$\xrightarrow{R_2\leftrightarrow R_3} \left(\begin{array}{ccc|c} 1&0&0&y_1\\ 0&1&0&y_3\\ 0&0&0&y_2-2y_1\\ 0&0&0&y_4-3y_1-4y_3\\ 0&0&1&y_5 \end{array}\right) \xrightarrow{R_3\leftrightarrow R_5} \left(\begin{array}{ccc|c} 1&0&0&y_1\\ 0&1&0&y_3\\ 0&0&1&y_5\\ 0&0&0&y_4-3y_1-4y_3\\ 0&0&0&y_2-2y_1 \end{array}\right)$$

Lecture 2, Proposition 7, with the unknowns c_1, c_2, c_3, $r = 3$, no free unknowns:

$$y \in W \iff y_2 - 2y_1 = 0,\ y_4 - 3y_1 - 4y_3 = 0 \iff y_2 = 2y_1,\ y_4 = 3y_1 + 4y_3 \quad ((a))$$
$$\text{then } c = (y_1, y_3, y_5)' :\ y = y_1v_1 + y_3v_2 + y_5v_3 \quad ((b))$$
$$(-3,-6,1,-5,2)' :\ -6 = 2(-3),\ -5 = 3(-3) + 4\cdot 1 \implies \in W,\ = -3v_1 + v_2 + 2v_3$$
$$(2,4,6,7,8)' :\ 3\cdot 2 + 4\cdot 6 = 30 \neq 7 \implies \notin W$$

Theorem 9. Let $A = [a_1\ \cdots\ a_n] \in \mathbb{R}^{m\times n}$ and $B = [b_1\ \cdots\ b_p] \in \mathbb{R}^{m\times p}$.

(a) $\operatorname{span}\{b_1, \ldots, b_p\} \subseteq \operatorname{span}\{a_1, \ldots, a_n\} \iff$ every $b_j \in \operatorname{span}\{a_1, \ldots, a_n\} \iff B = AF$ for some $F \in \mathbb{R}^{n\times p}$.

(b) $\operatorname{span}\{b_1, \ldots, b_p\} = \operatorname{span}\{a_1, \ldots, a_n\} \iff B = AF$ and $A = BG$ for some $F \in \mathbb{R}^{n\times p}$, $G \in \mathbb{R}^{p\times n}$.

(c) If every $b_j \in \operatorname{span}\{a_1, \ldots, a_n\}$, then $\operatorname{span}\{a_1, \ldots, a_n, b_1, \ldots, b_p\} = \operatorname{span}\{a_1, \ldots, a_n\}$. So deleting from a list a vector that is a linear combination of the others does not change the span.

Proof.

(a) $S_a = \operatorname{span}\{a_1, \ldots, a_n\}$, $S_b = \operatorname{span}\{b_1, \ldots, b_p\}$, $F = [f_1\ \cdots\ f_p]$:

$$S_b \subseteq S_a \implies b_j \in S_b \subseteq S_a \text{ (all } j) \qquad \text{(Theorem 7: } b_j \in S_b)$$
$$b_1, \ldots, b_p \in S_a \implies S_b \subseteq S_a \qquad \text{(Theorem 7: } S_a \text{ a subspace, } S_b \text{ the smallest)}$$
$$\operatorname{col}_j(AF) = Af_j = f_{1j}a_1 + \cdots + f_{nj}a_n \qquad \text{(Lecture 1, Theorem 4(a),(b))}$$

So $B = AF$ for some $F \iff$ each b_j is a linear combination of $a_1, \ldots, a_n$.

(b) The spans are equal $\iff$ each contains the other: (a) for both inclusions.

(c) $b_j = k_{1j}a_1 + \cdots + k_{nj}a_n$ $(j = 1, \ldots, p)$; the order of a list does not change its span:

$$\textstyle\sum_{i=1}^n x_ia_i + \sum_{j=1}^p y_jb_j = \sum_{i=1}^n \Big(x_i + \sum_{j=1}^p k_{ij}y_j\Big)a_i \in \operatorname{span}\{a_1, \ldots, a_n\}$$
$$\textstyle\sum_i x_ia_i = \sum_i x_ia_i + \sum_j 0\, b_j \in \operatorname{span}\{a_1, \ldots, a_n, b_1, \ldots, b_p\}$$
last sentence: a_i the remaining vectors, b_1 the deleted one $(p = 1)$

□

3. Linear independence

Definition 10. $v_1, \dots, v_k$ in a vector space are **linearly dependent** if $c_1v_1 + \cdots + c_kv_k = 0$ for some $c_1, \dots, c_k \in \mathbb{R}$, not all 0 (a **linear relation**); otherwise **linearly independent**: $c_1v_1 + \cdots + c_kv_k = 0$ only for $c_1 = \cdots = c_k = 0$. The empty list is independent.

Proposition 11. Let $v_1, \dots, v_k \in \mathbb{R}^m$, $V = [v_1 \ \cdots \ v_k] \in \mathbb{R}^{m\times k}$. Then $v_1, \dots, v_k$ are linearly independent $\iff$ $Vc = 0$ only for $c = 0$ $\iff$ every column of an RREF of V is a pivot column (no free unknowns). If $k > m$, then $v_1, \dots, v_k$ are linearly dependent.

Proof.

$Vc = c_1v_1 + \cdots + c_kv_k$ (Lecture 1, Theorem 4(b)):

linear relations among $v_1, \dots, v_k$ = solutions $c \neq 0$ of $Vc = 0$

$Vc = 0$ consistent $(c = 0)$: $c = 0$ only $\iff$ no free unknowns (Lecture 2, Theorem 22(b))

$k > m \implies Vc = 0$ for some $c \neq 0$ (Lecture 2, Theorem 9(a))

□

Proposition 12. Let $v_1, \dots, v_k$ be vectors of a vector space.

(a) If $v_i = 0$ for some i, or $v_i = v_j$ for some $i \neq j$, then $v_1, \dots, v_k$ are dependent. A single vector v is independent $\iff$ $v \neq 0$.

(b) A sublist of an independent list is independent.

(c) For $k \geqslant 2$: $v_1, \dots, v_k$ are dependent $\iff$ some v_j is a linear combination of the others. So two vectors are dependent $\iff$ one of them is a multiple of the other.

Proof.

(a) Relations with a coefficient 1, the other coefficients 0 (Proposition 2):

$$v_i = 0 : \ 1v_i = 0; \qquad v_i = v_j : \ 1v_i + (-1)v_j = 0$$

$$v \neq 0, \ cv = 0 \implies c = 0 \text{ (independent)}; \qquad v = 0 : \ 1v = 0 \text{ (dependent)} \qquad \text{(Proposition 2)}$$

(b) $S \subseteq \{1, \dots, k\}$ the indices of a sublist; a relation among the v_i, $i \in S$ (not all $c_i = 0$):

$$\textstyle\sum_{i\in S} c_iv_i = 0 \implies \sum_{i\in S} c_iv_i + \sum_{i\notin S} 0\, v_i = 0, \text{ a relation among all of them}$$

So a dependent sublist makes the whole list dependent; contrapositive: (b).

(c) Both directions; for two vectors, "the others" is the other one:

$$v_j = \textstyle\sum_{i\neq j} x_iv_i \implies v_j - \sum_{i\neq j} x_iv_i = 0, \text{ coefficient 1 on } v_j$$

$$\textstyle\sum_i x_iv_i = 0, \ x_j \neq 0 \implies v_j = \sum_{i\neq j}(-x_i/x_j)v_i$$

□

Example 13. Let $a_1 = (1,1,2)'$, $a_2 = (1,1,1)'$, $a_3 = (1,2,5)'$, $a_4 = (-2,-3,-4)'$, the columns of the matrix A of Lecture 2, Example 6. Find a linear relation among $a_1, \dots, a_4$; show that a_1, a_2, a_3 are independent.

Solution.

Four vectors of $\mathbb{R}^3$ are dependent (Proposition 11); A, R, $Ax = 0$ as restated:

$$\begin{aligned} x = (-2,3,1,1)' :\ & -2a_1 + 3a_2 + a_3 + a_4 = Ax = 0 && \text{(Lecture 1, Theorem 4(b))} \\ \implies\ & a_4 = 2a_1 - 3a_2 - a_3 \\ [a_1\ a_2\ a_3] \to\ & \text{columns 1–3 of } R, \text{ which form } I_3 && \text{(Lecture 2, Theorem 14(c))} \\ \implies\ & \text{all pivot columns} \implies a_1, a_2, a_3 \text{ independent} && \text{(Proposition 11)} \end{aligned}$$

4. Bases

Definition 14. $v_1, \dots, v_k$ in a vector space V form a **basis** of a subspace W if they lie in W, are linearly independent, and span W.

- $e_1, \dots, e_n$ form a basis of $\mathbb{R}^n$, the **standard basis**: $x = x_1e_1 + \cdots + x_ne_n$ (Lecture 1, Theorem 4(b) with I_n), and this is 0 only for $x = 0$.
- $E_{ij} \in \mathbb{R}^{m\times n}$ (entry (i,j) is 1, all others 0): the mn matrices E_{ij} form a basis of $\mathbb{R}^{m\times n}$, as $A = \sum_i \sum_j a_{ij}E_{ij}$, and $\sum_i \sum_j c_{ij}E_{ij} = (c_{ij})$ is 0 only if every $c_{ij} = 0$.
- The empty list is a basis of $\{0\}$.

Theorem 15. If $v_1, \dots, v_k$ form a basis of W, each $w \in W$ is $w = c_1v_1 + \cdots + c_kv_k$ for exactly one $(c_1, \dots, c_k)$: the **coordinates** of w in this basis.

Proof.

Existence: $v_1, \dots, v_k$ span W. Uniqueness:

$$\textstyle\sum_i x_iv_i = \sum_i y_iv_i \implies \sum_i (x_i - y_i)v_i = 0 \implies x_i = y_i \text{ (all } i) \qquad \text{(independence)}$$

□

Example 16. If $P \in \mathbb{R}^{n\times n}$ is invertible, its columns $p_1, \dots, p_n$ form a basis of $\mathbb{R}^n$, and the coordinates of $y \in \mathbb{R}^n$ are $x = P^{-1}y$. For the invertible A of Lecture 2, Example 19: find the coordinates of $y = (1,0,3,3)'$ in the basis of the columns of A.

Solution.

General: $Px = x_1p_1 + \cdots + x_np_n$ (Lecture 1, Theorem 4(b)):

$$\begin{aligned} Px = 0 \implies x = 0 :\ & p_1, \dots, p_n \text{ independent} && \text{(Lecture 2, Theorem 15)} \\ y = P(P^{-1}y) = (P^{-1}y)_1p_1 + \cdots + (P^{-1}y)_np_n & && (PP^{-1} = I_n\text{; Lecture 1, Theorem 3(a)}) \\ \text{they span } \mathbb{R}^n\text{; coordinates of } y\text{: } & P^{-1}y && \text{(Theorem 15)} \end{aligned}$$

Numbers: A, A^{-1} as restated:

$$\begin{aligned} A^{-1}y &= (0+0+3+0,\ -1+0+3-3,\ -1+0+0+0,\ -1+0+0+3)' = (3,-1,-1,2)' \\ y &= 3(1,-1,2,1)' - (-1,1,-1,-1)' - (1,-2,2,1)' + 2(-1,1,-1,0)' \end{aligned}$$

$A^{-1}y$ is the solution of $Ax = y$ found in Lecture 2, Example 23.

Example 17. In $W_0 = \{x \in \mathbb{R}^4 : x_1+x_2+x_3+x_4 = 0\}$, show that $w_1 = (1,-1,0,0)'$, $w_2 = (0,1,-1,0)'$, $w_3 = (0,0,1,-1)'$ form a basis.

Solution.

In W_0: the entries of each w_i add up to 0. Span and independence:

$$\begin{aligned} x \in W_0 &\implies x_4 = -x_1 - x_2 - x_3 \\ x_1w_1 + (x_1+x_2)w_2 + (x_1+x_2+x_3)w_3 &= (x_1,\ x_2,\ x_3,\ -x_1-x_2-x_3)' = x \\ c_1w_1 + c_2w_2 + c_3w_3 &= (c_1,\ c_2-c_1,\ c_3-c_2,\ -c_3)' = 0 \\ &\implies c_1 = 0,\ c_2 = 0,\ c_3 = 0 \end{aligned}$$

5. Sums and direct sums

Definition 18. For subspaces U, W of a vector space V: the **intersection** $U \cap W$ is the set of the vectors in both U and W; the **sum** is $U + W = \{u + w : u \in U,\ w \in W\}$. The sum is **direct**, written $U \oplus W$, if $U \cap W = \{0\}$. So $V = U \oplus W$ means $V = U + W$ and $U \cap W = \{0\}$.

Theorem 19. Let U, W be subspaces of a vector space V.

(a) $U \cap W$ and $U + W$ are subspaces of V.

(b) $U + W$ contains U and W, and it is contained in every subspace of V that contains $U \cup W$: it is the smallest such subspace.

(c) If $U = \operatorname{span}\{u_1, \dots, u_p\}$ and $W = \operatorname{span}\{w_1, \dots, w_q\}$, then $U+W = \operatorname{span}\{u_1, \dots, u_p, w_1, \dots, w_q\}$.

Proof.

(a) Theorem 4(c) for both; $c \in \mathbb{R}$, $u_1, u_2 \in U$, $w_1, w_2 \in W$:

$$\begin{aligned} 0 \in U,\ 0 \in W; \quad 0 = 0+0 \in U+W & \qquad ((\text{V3})) \\ x, y \in U \cap W \implies cx+y \in U,\ cx+y \in W \implies cx+y \in U \cap W & \qquad (\text{Theorem 4(c)}) \\ x = u_1+w_1,\ y = u_2+w_2 \implies cx+y = (cu_1+u_2)+(cw_1+w_2) \in U+W & \qquad ((\text{V1}), (\text{V2}), (\text{V7})) \end{aligned}$$

(b) Z a subspace of V containing $U \cup W$:

$$\begin{aligned} u = u+0,\ w = 0+w \implies U \subseteq U+W,\ W \subseteq U+W & \qquad ((\text{V1}), (\text{V3})) \\ u \in U,\ w \in W \implies u, w \in Z \implies u+w \in Z \implies U+W \subseteq Z & \qquad (\text{Theorem 4(b)}) \end{aligned}$$

(c) Both inclusions, with $S = \operatorname{span}\{u_1, \dots, u_p, w_1, \dots, w_q\}$:

$$\begin{aligned} u = \textstyle\sum_i a_iu_i \in U,\ w = \sum_j b_jw_j \in W \implies u+w = \sum_i a_iu_i + \sum_j b_jw_j \in S \\ \textstyle\sum_i a_iu_i + \sum_j b_jw_j = u+w, \quad u = \sum_i a_iu_i \in U,\ w = \sum_j b_jw_j \in W \end{aligned}$$

□

For $A = [a_1\ \cdots\ a_p] \in \mathbb{R}^{m\times p}$, $B = [b_1\ \cdots\ b_q] \in \mathbb{R}^{m\times q}$ and $[A\ B] \in \mathbb{R}^{m\times(p+q)}$ with the columns $a_1, \dots, a_p, b_1, \dots, b_q$: by (c) and Lecture 1, Theorem 4(b),

$$\operatorname{span}\{a_1, \dots, a_p, b_1, \dots, b_q\} = \operatorname{span}\{a_1, \dots, a_p\} + \operatorname{span}\{b_1, \dots, b_q\} = \{Ax + By : x \in \mathbb{R}^p,\ y \in \mathbb{R}^q\}$$

Theorem 20. Let U, W be subspaces of a vector space V. Then $U \cap W = \{0\} \iff$ each $x \in U+W$ is $x = u + w$ ($u \in U$, $w \in W$) in exactly one way: $u + w = u_1 + w_1$ with $u, u_1 \in U$, $w, w_1 \in W \Rightarrow u = u_1$, $w = w_1$.

Proof.

($\Rightarrow$) with $U \cap W = \{0\}$; ($\Leftarrow$) by contraposition:

$$\begin{aligned}
(\Rightarrow)\ u + w = u_1 + w_1 &\implies u - u_1 = w_1 - w && \text{((V1)–(V4))}\\
&\implies u - u_1 = w_1 - w \in U \cap W = \{0\},\ u = u_1,\ w = w_1 && \text{(Theorem 4(c))}\\
(\Leftarrow)\ 0 \neq x \in U \cap W &\implies x = x + 0 = 0 + x\ (x, 0 \in U;\ 0, x \in W):\ \text{two ways}
\end{aligned}$$

□

So, if $V = U \oplus W$, each $v \in V$ is $u + w$ for exactly one $u \in U$, $w \in W$ (the parts of v in U, W).

Theorem 21. Let U, W be subspaces of a vector space V, $u_1, \ldots, u_p$ a basis of U, $w_1, \ldots, w_q$ a basis of W. If $U \cap W = \{0\}$, then $u_1, \ldots, u_p, w_1, \ldots, w_q$ form a basis of $U \oplus W$. If $U \cap W \neq \{0\}$, then $u_1, \ldots, u_p, w_1, \ldots, w_q$ are linearly dependent. So $U + W$ is direct $\iff$ $u_1, \ldots, u_p, w_1, \ldots, w_q$ are linearly independent.

Proof.

Span: Theorem 19(b),(c); the two cases:

$$\begin{aligned}
U \cap W = \{0\}:\ \textstyle\sum_i a_i u_i + \sum_j b_j w_j = 0 &\implies x = \textstyle\sum_i a_i u_i = \sum_j (-b_j) w_j \in U \cap W\\
&\implies \textstyle\sum_i a_i u_i = x = 0,\ \sum_j b_j w_j = 0 \implies a_i = 0,\ b_j = 0:\ \text{a basis} && \text{(independence)}\\
0 \neq x \in U \cap W:\ x = \textstyle\sum_i a_i u_i = \sum_j b_j w_j &\implies \textstyle\sum_i a_i u_i + \sum_j (-b_j) w_j = 0\\
&\text{not all } a_i = 0\ (x \neq 0):\ \text{dependent; so direct} \iff \text{independent}
\end{aligned}$$

□

Example 22. Show that $\mathbb{R}^4 = W_0 \oplus \text{span}\{e_4\}$, with W_0 and its basis w_1, w_2, w_3 of Example 17; split $(1, 2, 3, 6)'$; show that w_1, w_2, w_3, e_4 form a basis of $\mathbb{R}^4$.

Solution.

Intersection, sum, and the basis ($e_4 \neq 0$ is a basis of $\text{span}\{e_4\}$: Proposition 12(a)):

$$\begin{aligned}
ce_4 = (0, 0, 0, c)' \in W_0 &\iff c = 0: \quad W_0 \cap \text{span}\{e_4\} = \{0\}\\
x - ce_4 = (x_1,\ x_2,\ x_3,\ x_4 - c)' \in W_0 &\iff c = x_1 + x_2 + x_3 + x_4\\
x = (x_1,\ x_2,\ x_3,\ -x_1 - x_2 - x_3)' + (x_1 + x_2 + x_3 + x_4)e_4 &\quad \text{(the only way)}\\
(1, 2, 3, 6)' = (1, 2, 3, -6)' + 12e_4; \quad w_1, w_2, w_3, e_4 \text{ form a basis of } \mathbb{R}^4 &\quad \text{(Theorem 21)}
\end{aligned}$$

Example 23. Show that $\mathbb{R}^n = \text{span}\{\mathbf{1}_n\} \oplus W_0$, $x = \bar{x}\mathbf{1} + Cx$; split $(1, 3, 5, 7, 9)'$. For $n = 4$, show that $\mathbf{1}_4, w_1, w_2, w_3$ form a basis of $\mathbb{R}^4$; compare with Example 22.

Solution.

Example 5(c); Lecture 1, Definition 14, Proposition 15, Example 16, Theorem 3(a),(b):

$$\begin{aligned}
x = \bar{x}\mathbf{1} + Cx, \quad \bar{x}\mathbf{1} \in \text{span}\{\mathbf{1}\},\ Cx \in W_0: &\quad \mathbb{R}^n = \text{span}\{\mathbf{1}\} + W_0\\
c\mathbf{1} \in W_0 \implies 0 = \mathbf{1}'(c\mathbf{1}) = cn,\ c = 0: &\quad \text{span}\{\mathbf{1}\} \cap W_0 = \{0\}\\
x = \bar{J}_n x + C_n x, \quad \bar{J}_n x = \tfrac{1}{n}\mathbf{1}(\mathbf{1}'x) = \bar{x}\mathbf{1}, &\quad \bar{J}_n + C_n = I_n\\
x = (1, 3, 5, 7, 9)':\ \bar{x} = 25/5 = 5, &\quad x = 5\mathbf{1} + (-4, -2, 0, 2, 4)'
\end{aligned}$$

So $\bar{x}\mathbf{1}$ is the only constant vector whose difference from x is centered (Theorem 20).

$n = 4$: $\mathbf{1}_4 \neq 0$ is a basis of $\operatorname{span}\{\mathbf{1}_4\}$ (Proposition 12(a)); Example 17:

$\mathbf{1}_4, w_1, w_2, w_3$ form a basis of $\mathbb{R}^4$ (Theorem 21)
$x = (1,2,3,6)' : \ \bar{x} = 3, \ Cx = (-2,-1,0,3)' \in W_0$; in Example 22: $(1,2,3,-6)' \in W_0$

The part in W_0 depends on the second subspace.

Example 24. In $\mathbb{R}^4$ let $U = \operatorname{span}\{u_1, u_2\}$, $W = \operatorname{span}\{w_1, w_2\}$, $u_1 = (1,0,1,2)'$, $u_2 = (1,1,0,1)'$, $w_1 = (0,1,0,1)'$, $w_2 = (2,0,1,2)'$. Find bases of $U + W$ and of $U \cap W$. Is the sum direct?

Solution.
$U + W = \operatorname{span}\{u_1, u_2, w_1, w_2\}$ (Theorem 19(c)); row-reduce $M = [u_1 \ u_2 \ w_1 \ w_2]$:

$$\begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 2 & 1 & 1 & 2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 0 & -1 \\ 0 & -1 & 1 & -2 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 + R_2]{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 2 & -2 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - 2R_3} \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_3} \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = R$$

Relations, $U + W$:

$$\begin{aligned} Mc = 0 &\iff Rc = 0 \iff c_1 + c_4 = 0, \ c_2 + c_4 = 0, \ c_3 - c_4 = 0 && \text{(Lecture 2, Theorem 3(b))} \\ &\iff c = t(-1,-1,1,1)' \ (c_4 = t \text{ free}) && \text{(Lecture 2, Proposition 7(b))} \\ t = 1 : \ & -u_1 - u_2 + w_1 + w_2 = 0, \quad w_2 = u_1 + u_2 - w_1 \\ U + W &= \operatorname{span}\{u_1, u_2, w_1\} && \text{(Theorem 9(c))} \\ [u_1 \ u_2 \ w_1] &\to \text{columns 1–3 of } R\text{: } e_1, e_2, e_3 && \text{(Lecture 2, Theorem 14(c))} \\ &\implies u_1, u_2, w_1 \text{ independent: a basis of } U + W && \text{(Proposition 11)} \end{aligned}$$

$U \cap W$, direct?

$$\begin{aligned} x \in U \cap W &\iff x = a_1 u_1 + a_2 u_2 = b_1 w_1 + b_2 w_2, \ M(a_1, a_2, -b_1, -b_2)' = 0 \text{ (some } a_i, b_i) \\ &\iff (a_1, a_2, -b_1, -b_2)' = t(-1,-1,1,1)', \ x = -t(u_1 + u_2) \text{ (some } t \in \mathbb{R}) \\ U \cap W &= \operatorname{span}\{u_1 + u_2\}, \quad u_1 + u_2 = (2,1,1,3)' = w_1 + w_2 \end{aligned}$$

$u_1 + u_2 \neq 0$ is a basis of $U \cap W$ (Proposition 12(a)); $U \cap W \neq \{0\}$, so the sum is not direct (Definition 18). u_1, u_2 are a basis of U (sublist of u_1, u_2, w_1, Proposition 12(b)), w_1, w_2 of W (not multiples of each other: first entries $0, 2$, second entries $1, 0$; Proposition 12(c)); accordingly u_1, u_2, w_1, w_2 are dependent (Theorem 21).

Practice. The Exercises and Problems of the textbook, Chapter 3: subspaces, span, linear independence, bases, and sums and direct sums of subspaces.